PETROCHEMICAL TRADE ANALYTICS

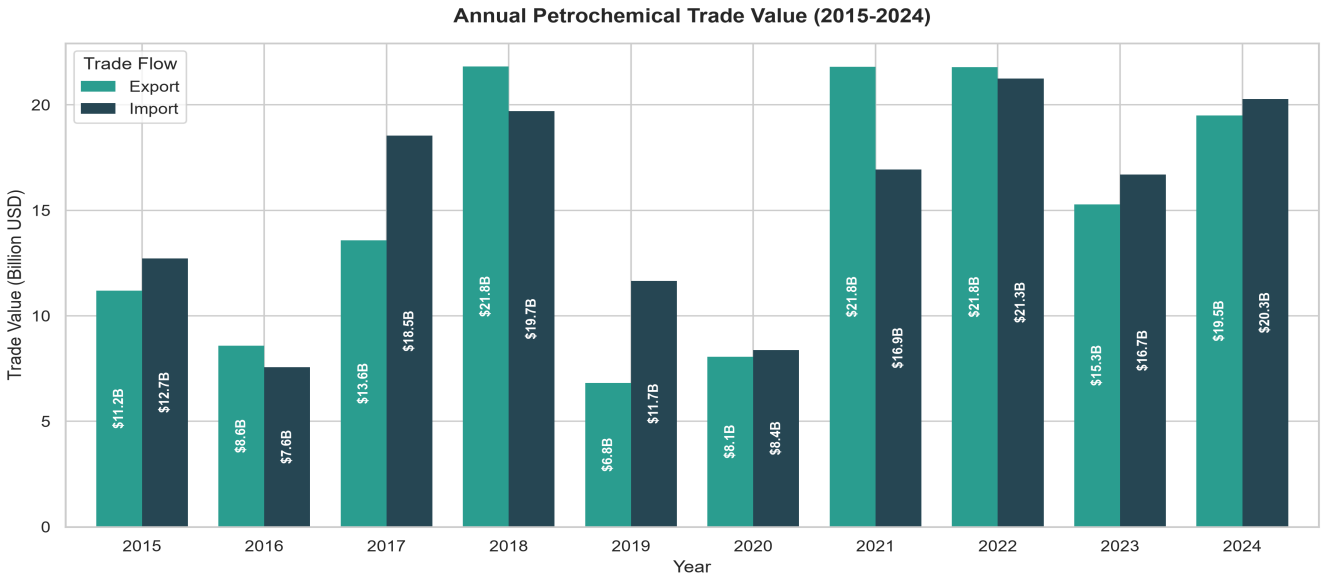
Global Trade Flows, Statistical Inference, Predictive Machine Learning, and Power BI Architecture
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1. Executive Summary & Core KPIs

This project analyzes 10 years (2015–2024) of UN Comtrade commercial transactions across five core petrochemical commodities (Ethylene, Propylene, Butadiene, Benzene, Toluene) and eight major economies (China, Germany, India, Japan, Netherlands, Rep. of Korea, Saudi Arabia, and the United States). Cumulative gross commercial value totaled **$302.18 Billion USD** across **334.13 Million Metric Tons**, with an aggregate net trade balance of **-$5.29 Billion USD**.

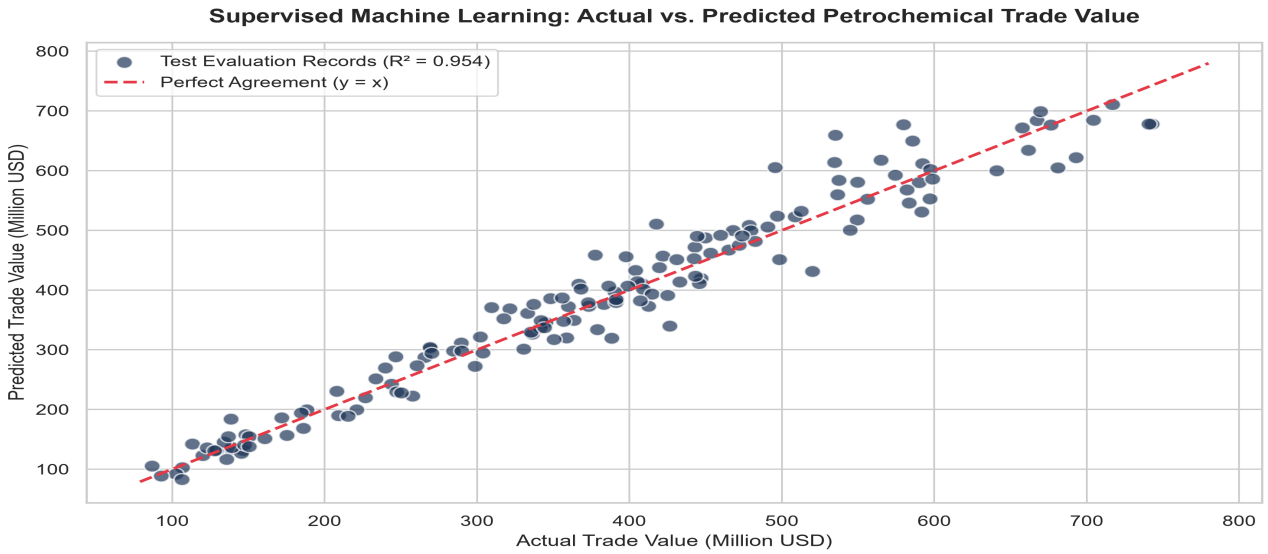
Total Trade Value	Total Imports	Total Exports	Trade Balance	Quantity Shipped	Data Reliability
$302.18B	$153.74B	$148.44B	-$5.29B	334.13 Mt	97.1% Usable

2. Annual Trade Trends (2015–2024)



3. Supervised Machine Learning Benchmark (Trade Forecasting)

Algorithm	Target Variable	MAE (USD)	RMSE (USD)	R² Score	MAPE (%)
Linear Regression	Trade_Value_USD	$39.21M	$50.14M	0.9069	11.06%
Ridge Regression (Alpha=1.0)	Trade_Value_USD	$39.20M	$50.17M	0.9068	11.04%
Random Forest Regressor	Trade_Value_USD	$28.37M	$37.77M	0.9471	8.19%
Gradient Boosting Regressor	Trade_Value_USD	$26.41M	$35.12M	0.9543	7.52%



4. Statistical Hypothesis Testing & Inference

Test ID	Test Name	Null Hypothesis (H0)	P-Value	Decision
HYP-01	Welch's Two-Sample T-Test	Mean Unit Price Pre-2020 == Mean Unit Price Post-2020	1.4236e-06	Reject H0 (Statistically Significant)
HYP-02	One-Way ANOVA	Mean prices across all 5 basic petrochemicals are equal	2.7026e-09	Reject H0 (Statistically Significant)
HYP-03	Pearson & Spearman Correlation	No correlation between transaction volume (MT) and unit price (USD/MT)	0.6757	Fail to Reject H0
HYP-04	Chi-Square Test of Independence	Commodity trade selection is independent of trade flow direction	0.5137	Fail to Reject H0

5. Relational Star Schema & Power BI Enterprise Architecture

The project implements a full Star Schema database design in SQLite (`petrochemical_trade.db`) and CSV tables ready for Power BI Desktop. The dimensional model separates commercial transactions into **Fact_TradeTransactions** connected via primary/foreign key pairs to **Dim_Commodity**, **Dim_Country**, and **Dim_Date**. Advanced DAX expressions compute Total Trade, Net Balance, Coverage Ratio, and Year-over-Year growth dynamically across all filter slices.